CCC–C–419b
July 9, 1964
SUPERSEDING
Fed. Spec. CCC–C–419a
April 26, 1961

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON, UNBLEACHED, PLIED-YARNS, ARMY AND NUMBERED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1. Scope. This specification covers the requirements of grey (unbleached) cotton duck cloth.

1.2 Classification.

1.2.1 *Types.* The cloth, cotton, duck, plied-yarns (Army and Numbered) shall be of the following types as specified (see 6.2):

Type I—Numbered duck.
Type II—(see 6.4).
Type III—Army Duck.
Type IV—(see 6.4).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V–T–276—Thread, Cotton.
CCC–T–191—Textile Test Methods.
PPP–P–51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Yarn.* Yarns shall be made from cotton which has been cleaned, carded, drawn, spun, and twisted into plied yarns.

3.2 Physical requirements. Cloth covered by this specification shall be as specified (see 6.2) and shall conform to the requirements specified in tables I and II when tested as specified in 4.4.

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TABLE I.—*Physical requirements—type I, numbered duck*

Number of duck, commercial designation	Weight, per square yard, minimum	Yarn ply, minimum		Yarns per in., minimum		Breaking strength, minimum		Air permeability, maximum
		Warp	Filling	Warp	Filling	Warp	Filling	
Hard texture:	*Ounces*					*Pounds*	*Pounds*	*Ft³/min/ft²*
12/0	48. 00	9	9	20	12	800	430	
8/0	40. 00	8	14	26. 5	12. 5	750	600	
2/0	31. 90	5	7	24	16	465	435	
1/0	30. 31	5	6	24	16	450	405	
1	28. 71	5	5	26	19	440	370	2
2	27. 12	5	4	26	19	420	345	
3	25. 53	4	5	29	20	390	330	
4	23. 93	4	4	31	22	375	300	2
5	22. 33	3	4	35	23	345	285	
6	20. 74	3	3	35	25	335	250	2
8	17. 55	3	3	43	25	285	210	2
10	14. 35	3	3	45	27	245	160	4
11	12. 75	3	2	45	34	240	140	
12	11. 16	2	2	45	35	195	120	4
Medium texture:								
1	28. 71	5	5	26	15	425	345	
2	27. 12	5	4	26	15	410	320	
3	25. 53	4	5	29	15	370	315	
4	23. 93	4	4	29	17	350	290	
5	22. 33	3	4	34	18	315	285	
6	20. 74	3	3	34	18	305	250	

TABLE II.—*Physical requirements—type III, Army duck*

Weight per square yard, minimum	Yarn ply, minimum		Yarns per inch, minimum		Breaking strength, minimum		Air permeability, maximum
	Warp	Filling	Warp	Filling	Warp	Filling	
Ounces					*Pounds*	*Pounds*	*Ft³/min./ft²*
8.25	2	2	60	58	125	120	
9.85	2	2	52	40	160	110	4
12.29	3	2	44	34	210	130	4
14.77	3	3	44	28	235	175	
15.90	3	3	45	34	245	200	
18.48	4	4	42	28	315	200	

3.2.1 *Width.* The width of the cloth inclusive of selvages shall be as specified (see 6.2). The following tolerances shall be permitted.

Width of duck (inches)	Tolerance (inch)
40 or less	+¼–¼
Over 40 to 51 inclusive	+⅜–¼
Over 51 to 81 inclusive	+⅜–⅜
Over 81	+¾–⅜

3.2.2 *Weave.* The weave shall be plain, one up, one down.

3.3 Nonfibrous material content. The cloth shall contain not more than 2.5 percent starch and protein content excluding chloroform-soluble and water-soluble material. The total sizing, finishing, and other nonfibrous material content of the finished cloth shall not exceed 5.0 percent when tested as specified in 4.4. The use of finishing or loading materials

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to increase the weight or breaking strength is prohibited and the supplier shall furnish a certificate of compliance with respect to this requirement.

3.4 Length and put-up. The cloth shall be put-up in rolls in accordance with PPP–P–51. Pieces 80 yards or more in length shall be delivered. When specified (see 6.2 and 6.3) pieces less than 80 yards will be acceptable up to 10 percent of contract quantity, provided no piece is less than 20 yards in length and not more than two pieces less than 80 yards are included in any one roll. For duck No. 10 and lighter in weight, pieces shall be sewn together with a butted FSa–1 type seam of Fed. Std. No. 751 using ticket No. 10, 3 ply unbleached cotton thread, conforming to type IIIA of V–T–276, and having 6 to 8 stitches per inch. The stitching shall be placed $\frac{1}{2} \pm 1/16$ inch from the cut or raw edge of the cloth. For cloth heavier in weight than No. 10, the pieces shall be lapped and not sewn together. The number of pieces in the roll shall be indicated by stencil marking on the last 6 inches of the roll on the outer surface.

3.5 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.6 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable point level specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL–STD–105 and 4.2.1 through 4.2.3.

4.2.1 *Yard-by-yard examination.* The required yardage of each roll shall be examined on the face side and all defects as defined in Section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except only those slubs and knots which exceed the maximum acceptable limits shown for each in figure 1 shall be scored. No one of the 30 contiguous yards (increments of 1 yard on the measuring device) selected for examination from any one roll within the sample shall be penalized more than 4 points.

The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 rolls. An approximately equal number of yards shall be examined in each roll of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 40.00. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{Contracted width of material (inches)}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

	Points
For defects up to 3 inches in any dimension	1
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	2
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	3
For defects exceeding 9 inches in any dimension	4

NOTE.—The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Width not within specified tolerance.
Objectionable odor.
Excessive neppiness.

4.2.2 *Examination for length.*

4.2.2.1 *Individual rolls.* During the yard-by-yard examination each roll in the sample shall be examined for put-up and length. Each defect shall be counted not more than once in each roll examined. The lot shall be unacceptable if two or more of the following defects are present:

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Defects

Any length found to be more than 2 yards below the length marked on the roll ticket.
Length of piece:
Regular lengths—Any piece less than 80 yards in length.
Short lengths—(When applicable) any piece less than 20 yards in length or more than two pieces less than 80 yards in length in one roll.
For No. 10 duck and lighter in weight:
(*a*) Pieces not sewn together with a butted FSa–1 type seam.
(*b*) 6 to 8 stitches per inch not used in seam.
(*c*) Stitching not $\frac{1}{2} \pm \frac{1}{16}$ inch from the cut or raw edge of the cloth.
For duck heavier in weight than No. 10, pieces sewn instead of lapped.
Number of pieces in roll not stenciled on last 6 inches of rolls on outer surface.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the ticket.

4.2.3 *Examination for compliance with Textile Fiber Products Identification Act.* During the yard-by-yard examination, each roll in the sample shall be examined for compliance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP–P–51 to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

4.4 Testing of the end item. The methods of testing specified in CCC–T–191 wherever applicable as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1½ yards full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirements specified. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method
Weight	Tables I and II	5041.[1]
Weave	3.2.2	Visual.[2]
Breaking strength	Tables I and II	5100.
Yarn ply	Tables I and II	Visual.[2]
Yarns per inch	Tables I and II	5050.[3]
Starch and protein content.	3.3	2611.
Total sizing, finishing and other nonfibrous materials.	3.3	2611.
Air permeability	Tables I and II	5450.

[1] Except that the average of 5 determinations per sample unit shall be reported to the nearest 0.01 ounce.
[2] One determination shall be made from each sample unit and the result reported as "pass" or "fail".
[3] Except that for type I, 8/0 H.T., the results shall be reported to the nearest 0.1 of a yarn.

5. PREPARATION FOR DELIVERY

(For Civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A and C.* The cloth shall be put up as specified and packaged in accordance with the applicable requirements for class 4 fabrics of PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2 and 5.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP–P–51.

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6. NOTES

6.1 Intended use. The cloths covered by this specification are primarily intended for use in fabrication of tentage and equipage items.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type of cloth required (see 1.2).
(*c*) Width required (see 3.2.1).
(*d*) Acceptance of pieces less than 80 yards (see 3.4).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives as indicated in Specification PPP–P–51 may be exercised in the preparation for delivery of cloth.

6.3 Short pieces. Pieces less than 80 yards in length in excess of the amount specified in 3.4 and pieces shorter in length than 25 yards may be presented to the contracting officer for consideration for end uses where short length pieces may be considered.

6.4 Types II and IV have been deleted and will not be required or cited for procurement.

6.5 Transportation description. The transportation description and minimum weights applicable to the commodity are:

Rail:
Cotton piece goods, not otherwise indexed by name, in the original piece.
Carload minimum weight—any quantity.

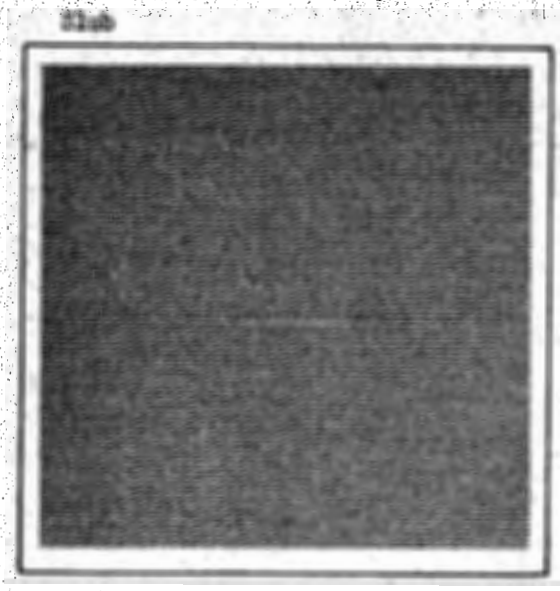
Motor:
Cotton piece goods, not otherwise indexed, in the original piece.
Truckload minimum weight—any quantity.

6.6 When level B packaging requirements are specified for civil agency procurements, the requirements in paragraph 5.1.1 shall apply.

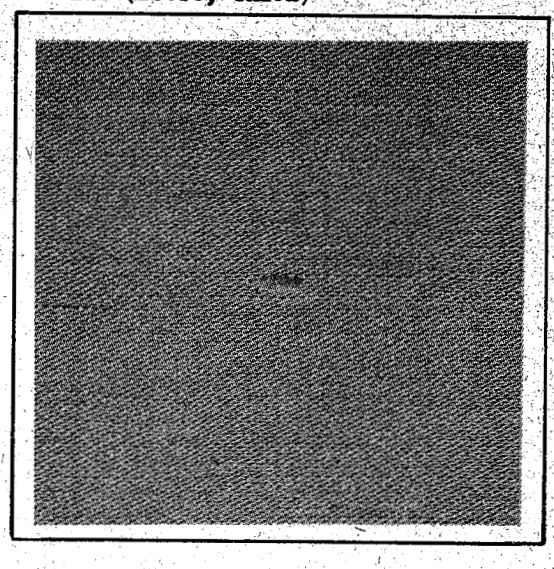
CUSTODIANS:
Army—GL
Navy—SA
Air Force—11
Preparing activity:
Army—GL
Interested activities:
Review
Army—MI, MD, MU
Navy—SA, MC
Air Force—11, 67
DSA—DM
User
Army—MR, EL, MO
Navy—YD, SH, CG

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Slub (Loose, Thick)



Lump or Knot (Fuzz)

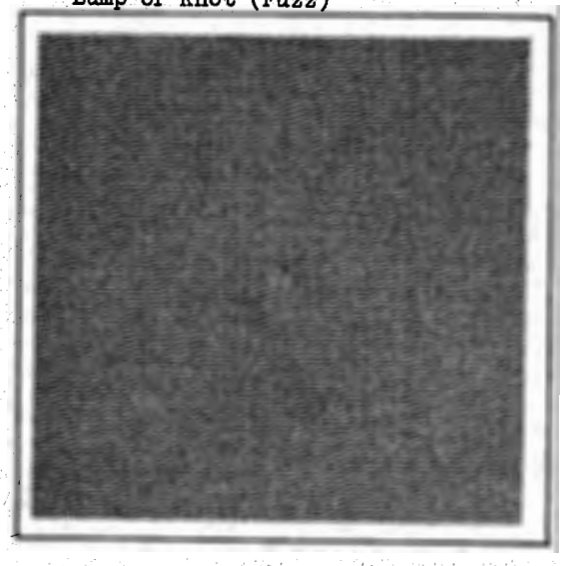


FIGURE 1.—(*Limits of Acceptability.*)

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